

Project Design Phase-II

Data Flow Diagram & User Stories

Date	31 January 2025
Team ID	Individual Project
Project Name	Book - A - Doctor
Maximum Marks	4 Marks

Data Flow Diagram (DFD)

Level-0 DFD (Context Diagram)

External Entities:

- Patient
- Doctor

Main Process:

- Book A Doctor System

Data Flow:

1. Patient registers/logs in to the system.
2. Patient searches doctors and books appointments.
3. Patient uploads medical reports.
4. Doctor logs in to the system.
5. Doctor views appointment requests.
6. Doctor approves/rejects appointments.
7. System stores and retrieves data from MongoDB.
8. Appointment status and reports are displayed to users.

DFD Description

Patient → Book A Doctor System

- Registration Details
- Login Credentials
- Appointment Requests
- Medical Reports

Book A Doctor System → Patient

- Authentication Response
- Doctor Information
- Appointment Status
- Dashboard Information

Doctor → Book A Doctor System

- Login Credentials
- Profile Information
- Appointment Approval/Rejection

Book A Doctor System → Doctor

- Appointment Requests
- Patient Information
- Dashboard Data

Book A Doctor System ↔ MongoDB Database

- User Data
- Doctor Data
- Appointment Data
- Medical Reports Data

User Stories

User Type: Patient

Epic: Registration

USN-1

- User Story: As a patient, I can register by entering my personal details, email, and password.
- Acceptance Criteria: Patient account is created successfully and stored in the database.
- Priority: High
- Release: Sprint-1

USN-2

- User Story: As a patient, I can log in using my registered email and password.
- Acceptance Criteria: Patient is redirected to the dashboard after successful login.
- Priority: High

- Release: Sprint-1

Epic: Doctor Search

USN-3

- User Story: As a patient, I can view the list of available doctors.
- Acceptance Criteria: All registered doctors are displayed successfully.
- Priority: High
- Release: Sprint-2

USN-4

- User Story: As a patient, I can search doctors based on specialization.
- Acceptance Criteria: Matching doctors are displayed according to search criteria.
- Priority: High
- Release: Sprint-2

Epic: Appointment Booking

USN-5

- User Story: As a patient, I can book an appointment with a doctor.
- Acceptance Criteria: Appointment request is created and stored successfully.
- Priority: High
- Release: Sprint-3

USN-6

- User Story: As a patient, I can view appointment status.
- Acceptance Criteria: Appointment status is displayed correctly.
- Priority: High
- Release: Sprint-3

USN-7

- User Story: As a patient, I can cancel my appointment.
- Acceptance Criteria: Appointment status changes to cancelled.
- Priority: Medium
- Release: Sprint-3

Epic: Medical Reports

USN-8

- User Story: As a patient, I can upload medical reports.
- Acceptance Criteria: Medical report is uploaded and linked to my profile.
- Priority: High
- Release: Sprint-4

USN-9

- User Story: As a patient, I can view uploaded medical reports.
- Acceptance Criteria: Uploaded reports are displayed successfully.
- Priority: Medium
- Release: Sprint-4

User Type: Doctor

Epic: Doctor Management

USN-10

- User Story: As a doctor, I can register and create my profile.
- Acceptance Criteria: Doctor profile is created successfully.
- Priority: High
- Release: Sprint-1

USN-11

- User Story: As a doctor, I can log in securely.
- Acceptance Criteria: Doctor dashboard is displayed after successful login.
- Priority: High
- Release: Sprint-1

Epic: Appointment Management

USN-12

- User Story: As a doctor, I can view appointment requests.
- Acceptance Criteria: All appointment requests are displayed.
- Priority: High
- Release: Sprint-3

USN-13

- User Story: As a doctor, I can approve appointment requests.
- Acceptance Criteria: Appointment status changes to Approved.
- Priority: High
- Release: Sprint-3

USN-14

- User Story: As a doctor, I can reject appointment requests.
- Acceptance Criteria: Appointment status changes to Rejected.
- Priority: High
- Release: Sprint-3

Epic: Dashboard

USN-15

- User Story: As a doctor, I can view my scheduled appointments.
- Acceptance Criteria: Upcoming appointments are displayed correctly.
- Priority: High
- Release: Sprint-4

USN-16

- User Story: As a doctor, I can access patient details and reports.
- Acceptance Criteria: Patient information and uploaded reports are visible.
- Priority: Medium
- Release: Sprint-4

User Type: Administrator/System

Epic: Security & Access Control

USN-17

- User Story: As a system, I must authenticate users using JWT.
- Acceptance Criteria: Only authenticated users can access protected resources.
- Priority: High
- Release: Sprint-1

USN-18

- User Story: As a system, I must provide role-based access control.
- Acceptance Criteria: Patients and doctors can only access authorized features.

- Priority: High
- Release: Sprint-4

USN-19

- User Story: As a system, I must securely store user and appointment data.
- Acceptance Criteria: Data is stored successfully in MongoDB and retrieved when required.
- Priority: High
- Release: Sprint-4